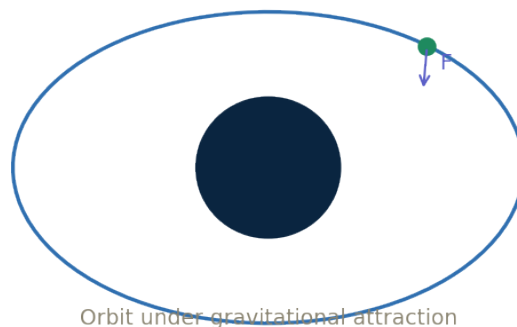


Gravitation

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Newton's Law of Gravitation: every particle attracts every other particle with a force $F = G \frac{m_1 m_2}{r^2}$, directed along the line joining them.
- Acceleration due to gravity $g = GM/R^2$ at Earth's surface. It decreases with height ($g' = g(1 - 2h/R)$ for $h \ll R$) and with depth ($g' = g(1 - d/R)$).
- Gravitational potential energy $U = -G \frac{M m}{r}$ (taken as zero at infinity). Escape velocity is the minimum speed to escape a planet's gravity permanently.
- Kepler's three laws describe planetary motion: (1) elliptical orbits with the Sun at one focus, (2) equal areas swept in equal times (areal velocity constant), (3) T^2 is proportional to a^3 (a = semi-major axis).
- Orbital velocity of a satellite close to Earth's surface: $v_o = \sqrt{GM/R}$. Escape velocity: $v_e = \sqrt{2GM/R} = \sqrt{2} \times v_o$.
- A geostationary satellite orbits in the equatorial plane with a time period equal to Earth's rotation period (24 hours), appearing stationary from the ground.



A satellite in elliptical orbit around Earth under gravitational attraction F .

2. Formula Sheet

Newton's law of gravitation	$F = G \frac{m_1 m_2}{r^2}$
Acceleration due to gravity	$g = \frac{GM}{R^2}$
g at height h ($h \ll R$)	$g' = g \left(1 - \frac{2h}{R}\right)$
g at depth d	$g' = g \left(1 - \frac{d}{R}\right)$
Gravitational PE	$U = -G \frac{M m}{r}$
Orbital velocity	$v_o = \sqrt{GM/R}$
Escape velocity	$v_e = \sqrt{2GM/R}$
Kepler's third law	$T^2 \text{ proportional to } r^3$

3. Practice Problems (22 questions)

- Find the gravitational force between two masses of 5 kg and 10 kg separated by 2 m ($G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$).
- Find the value of g at a height equal to the radius of the Earth above its surface (g on surface = 9.8 m/s^2).
- Find the value of g at a depth of $R/2$ below the Earth's surface.
- A satellite orbits close to the Earth's surface. Find its orbital speed, given $R(\text{earth}) = 6400 \text{ km}$ and $g = 9.8 \text{ m/s}^2$.
- Find the escape velocity from the Earth's surface using $R = 6400 \text{ km}$ and $g = 9.8 \text{ m/s}^2$.
- State Kepler's three laws of planetary motion.
- A planet has twice the mass and twice the radius of Earth. Find the acceleration due to gravity on its surface in terms of $g(\text{earth})$.
- Two satellites orbit the Earth at radii r and $4r$. Find the ratio of their time periods.
- Find the gravitational potential energy of a 10 kg object at the Earth's surface, given $M(\text{earth}) = 6 \times 10^{24} \text{ kg}$, $R(\text{earth}) = 6.4 \times 10^6 \text{ m}$.
- Explain why astronauts in an orbiting spacecraft experience weightlessness even though Earth's gravity is still acting on them.
- A geostationary satellite has a time period of 24 hours. State two conditions necessary for a satellite to be geostationary.
- If the mass of the Earth were doubled while its radius stayed the same, how would the value of g change?
- Find the ratio of escape velocity to orbital velocity for a satellite near a planet's surface.
- A body weighs 100 N on the surface of the Earth. Find its weight at a height equal to the Earth's radius.
- Derive the expression for orbital velocity of a satellite revolving close to a planet's surface.
- The mass of the Moon is about $1/81$ that of Earth and its radius is about $1/4$ that of Earth. Find g on the Moon's surface in terms of $g(\text{earth})$.

17. Explain why the value of g is zero at the center of the Earth.
18. A planet takes 8 times as long as Earth to complete one orbit around the Sun. Find the ratio of the planet's orbital radius to Earth's orbital radius using Kepler's third law.
19. Find the time period of a satellite orbiting at a height equal to Earth's radius above the surface (in terms of the time period T_0 of a satellite orbiting just above the surface).
20. State the difference between gravitational mass and inertial mass.
21. A body is projected vertically upward from the Earth's surface with half the escape velocity. Find the maximum height it reaches in terms of Earth's radius R .
22. Explain why the gravitational potential energy is taken to be negative in the formula $U = -GMm/r$.

Answer Key

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|---|--|
| 1. $F = 8.34 \times 10^{-10} \text{ N}$ | 2. $g' = g/4 = 2.45 \text{ m/s}^2$ |
| 3. $g' = g/2 = 4.9 \text{ m/s}^2$ | 4. $v_o \approx 7.9 \text{ km/s}$ |
| 5. $v_e \approx 11.2 \text{ km/s}$ | 6. Conceptual - see notes above |
| 7. $g' = g(\text{earth})/2$ | 8. $T \text{ ratio} = 1 : 8$ (since T^2 proportional to r^3) |
| 9. $U = -6.25 \times 10^8 \text{ J}$ (approx) | 10. Conceptual - both astronaut and spacecraft are in free fall, accelerating together |
| 11. Conceptual - orbit in equatorial plane, period = 24 h, same direction as Earth's rotation | 12. g doubles |
| 13. $\sqrt{2} : 1$ | 14. Weight = 25 N |
| 15. Conceptual derivation - see notes above | 16. $g(\text{moon}) \approx g/6$ |
| 17. Conceptual - net gravitational force from all directions cancels at the center | 18. Ratio = 4 : 1 (since $8^2 = r^3$ gives $r = 4$) |
| 19. $T = 2\sqrt{2} T_0$ (approx 2.83 T_0) | 20. Conceptual - gravitational mass measures gravitational force; inertial mass measures resistance to acceleration; experimentally found equal |
| 21. $H = R/3$ | 22. Conceptual - because PE is defined to be zero at infinity, and work must be done against attraction to reach infinity, so PE at finite r is negative |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.